Hexagon Number Pyramids

Year 4 · Homework Worksheet

Worked Example

Standard: A 4-hexagon base row has 3, 6, 2, 8. What is the top number?

Answer: Row 2: 3+6=9, 6+2=8, 2+8=10. Row 3: 9+8=17, 8+10=18. Top: 17+18=35.

Easy / Standard

(15 questions)

1. A 3-hexagon base row has the numbers 9, 2, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?
2. A 3-hexagon base row has the numbers 8, 9, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?
3. A 3-hexagon base row has the numbers 12, 8, 15. Each hexagon above is the sum of the two hexagons below it. What is the top number?
4. A 3-hexagon base row has the numbers 5, 9, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?
5. A 3-hexagon base row has the numbers 2, 2, 6. Each hexagon above is the sum of the two hexagons below it. What is the top number?
6. A 3-hexagon base row has the numbers 6, 5, 10. Each hexagon above is the sum of the two hexagons below it. What is the top number?
7. A 3-hexagon base row has the numbers 5, 14, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?
8. A 3-hexagon base row has the numbers 8, 15, 6. Each hexagon above is the sum of the two hexagons below it. What is the top number?
9. A 3-hexagon base row has the numbers 4, 7, 2. Each hexagon above is the sum of the two hexagons below it. What is the top number?
10. A 3-hexagon base row has the numbers 7, 11, 3. Each hexagon above is the sum of the two hexagons below it. What is the top number?
11. A 3-hexagon base row has the numbers 11, 8, 12. Each hexagon above is the sum of the two hexagons below it. What is the top number?
12. A 3-hexagon base row has the numbers 12, 15, 14. Each hexagon above is the sum of the two hexagons below it. What is the top number?
13. A 3-hexagon base row has the numbers 13, 13, 2. Each hexagon above is the sum of the two hexagons below it. What is the top number?

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14. A 3-hexagon base row has the numbers 9, 8, 3. Each hexagon above is the sum of the two hexagons below it. What is the top number?

15. A 3-hexagon base row has the numbers 8, 5, 15. Each hexagon above is the sum of the two hexagons below it. What is the top number?

Standard

(15 questions)

1. A 4-hexagon base row is 8, 13, ?, 18. The second row up reads 21, 25, 30 and the top hexagon is 101 (addition rule). What is the missing base number?

2. A 4-hexagon base row is 13, 16, ?, 9. The second row up reads 29, 35, 28 and the top hexagon is 127 (addition rule). What is the missing base number?

3. A 4-hexagon base row is 11, 13, ?, 18. The second row up reads 24, 28, 33 and the top hexagon is 113 (addition rule). What is the missing base number?

4. A 4-hexagon base row is 5, 11, ?, 4. The second row up reads 16, 20, 13 and the top hexagon is 69 (addition rule). What is the missing base number?

5. A 4-hexagon base row has 15, 7, 11, 4 (each hexagon above = sum of the two below). What is the top number? A) 66 B) 88 C) 78 D) 73

6. A 4-hexagon base row is 18, 15, ?, 9. The second row up reads 33, 30, 24 and the top hexagon is 117 (addition rule). What is the missing base number?

7. A 4-hexagon base row is 3, 9, ?, 3. The second row up reads 12, 17, 11 and the top hexagon is 57 (addition rule). What is the missing base number?

8. A 4-hexagon base row is 11, 6, ?, 15. The second row up reads 17, 21, 30 and the top hexagon is 89 (addition rule). What is the missing base number?

9. A 4-hexagon base row is 10, 20, ?, 9. The second row up reads 30, 24, 13 and the top hexagon is 91 (addition rule). What is the missing base number?

10. A 4-hexagon base row has 8, 13, 20, 18 (each hexagon above = sum of the two below). What is the top number? A) 125 B) 130 C) 112 D) 133

11. A 4-hexagon base row is 4, 6, ?, 20. The second row up reads 10, 24, 38 and the top hexagon is 96 (addition rule). What is the missing base number?

12. A 4-hexagon base row is 11, 7, ?, 14. The second row up reads 18, 11, 18 and the top hexagon is 58 (addition rule). What is the missing base number?

13. A 4-hexagon base row is 5, 19, ?, 12. The second row up reads 24, 22, 15 and the top hexagon is 83 (addition rule). What is the missing base number?

14. A 4-hexagon base row is 14, 5, ?, 20. The second row up reads 19, 10, 25 and the top hexagon is 64 (addition rule). What is the missing base number?

15. A 4-hexagon base row has 17, 15, 9, 12 (each hexagon above = sum of the two below). What is the top number? A) 106 B) 101 C) 86 D) 118

Standard / Advanced

(15 questions)

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1. A 5-hexagon base row has 16, 7, 6, 7, 23. Each hexagon above is the sum of the two below. What is the single top number?
2. A 3-hexagon base row has 16, 18 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 66. Find the missing base number.
3. Build the full pyramid from base row 17, 5, 9 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?
4. A 5-hexagon base row has 24, 13, 19, 17, 7. Each hexagon above is the sum of the two below. What is the single top number?
5. A 3-hexagon base row has 10, 16 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 50. Find the missing base number.
6. Build the full pyramid from base row 8, 7, 13 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?
7. A 5-hexagon base row has 19, 14, 12, 21, 4. Each hexagon above is the sum of the two below. What is the single top number?
8. A 3-hexagon base row has 10, 5 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 81. Find the missing base number.
9. Build the full pyramid from base row 12, 6, 6 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?
10. A 5-hexagon base row has 19, 23, 19, 20, 6. Each hexagon above is the sum of the two below. What is the single top number?
11. A 3-hexagon base row has 12, 18 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 58. Find the missing base number.
12. Build the full pyramid from base row 14, 10, 8 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?
13. A 5-hexagon base row has 24, 4, 25, 5, 23. Each hexagon above is the sum of the two below. What is the single top number?
14. A 3-hexagon base row has 15, 19 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 90. Find the missing base number.
15. Build the full pyramid from base row 12, 17, 17 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

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